



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

FULL SYLLABUS

14-08-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	A	D	B	B	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	C,D	B,C	A,B	A,B	A,B,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	20.00	0.4	4.00	0.4	440.00	3.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	B	C	D	D	D
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	B	A,B,C,D	A,B,C,D	B,C,D	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1.62	18.00	1.50	0.25	6.00	10.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B	A	D	A	D	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,C,D	A,B,D	A,B,C,D	A,C	A,B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	0.00	0.50	0.54	1.08	23.00	1.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

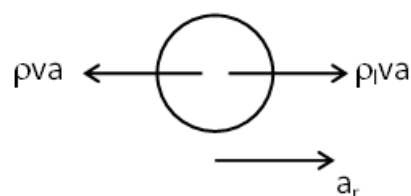
1. Ans (D)

$$\rho g 2x A + 2kx = Ma ; \quad \rho A 2L = M$$

$$a = \left(\frac{g}{L} + \frac{2k}{M} \right) x \Rightarrow \omega = \sqrt{\frac{g}{L} + \frac{2k}{M}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \left[\frac{g}{L} + \frac{2k}{M} \right]^{-1/2}$$

2. Ans (A)



$$N\ell = \frac{1}{2}at^2 \text{ (for container)}$$

$$\rho\ell va - \rho va = \rho va_r$$

$$a_r = \left(\frac{\rho\ell - 1}{\rho} \right) a$$

$$\ell = \frac{1}{2}a_r t^2$$

$$N = \frac{a}{a_r} = \frac{1}{\left(\frac{\rho\ell}{\rho} - 1 \right)} \Rightarrow \rho\ell = \left(\frac{1+N}{N} \right) \rho$$

3. **Ans (D)**

$$\tau = RC = 1 \text{ ms} \rightarrow (\ln 3); v_p - v_Q = e^{-\frac{t}{\tau}}$$

$$\Rightarrow \text{at time } (\ln 3) \text{ ms} \rightarrow V_p - V_Q = 2V$$

$$\text{Just after time } t = (\ln 3) \text{ ms} : V_p - V_Q = -4V$$

which then decreases to zero

Heat dissipated in resistor in time $0 \rightarrow (\ln 3) \text{ ms}$

$$= (6 \times 4 - 8) = 16 \mu\text{J}$$

Heat dissipated in resistor in time $(\ln 3) \text{ ms} \rightarrow \infty$

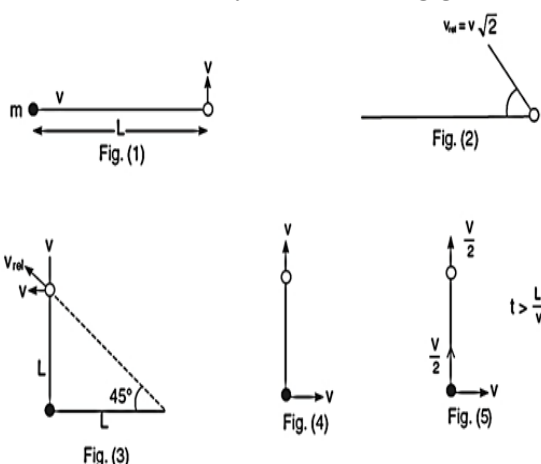
$$= \frac{1}{2} \times 1 \times 16 = 8 \mu\text{J}$$

4. **Ans (B)**

We will study the motion of second particle in the reference frame attached to the first particle. The velocity of second particle makes an angle of 45° with the initial line joining the two particles (see fig 2). The thread is loose before the distance between particles again becomes L . Fig.3 shows the situation just before the string gets taut.

$$\text{Required time is } t = \frac{L\sqrt{2}}{v\sqrt{2}} = \frac{L}{v}$$

In the reference frame of ground, velocities just before the string gets taut, has been shown in fig.4. The velocity component for the two particles along the string will be same for both particles after the string is taut. Fig.5 shows the situation immediately after the string gets taut.



Now, total K.E. =

$$\frac{1}{2} \frac{mv^2}{4} + \frac{1}{2} mv^2 \left(1^2 + \frac{1}{2^2} \right) = \frac{3}{4} mv^2$$

5. **Ans (B)**

Let amplitude of wave on arc m and p be a due to individual sources

$$\text{Amplitude on arc } \ell \text{ due to } S_1 = 2a$$

$$\text{Amplitude on arc } q \text{ due to } S_2 = \frac{a}{2}$$

At point A : $\Delta\phi =$

$$\frac{2\pi}{\lambda} \Delta x + \pi = \frac{2\pi}{4} (2) + \pi = 2\pi$$

$$\therefore a_A = \sqrt{a^2 + \left(\frac{a}{2}\right)^2 + 2.a.\frac{a}{2} \cdot \cos 2\pi} = \frac{\sqrt{5}a}{2}$$

$$\text{At B ; } \Delta\phi = \pi \quad [\because \text{no path difference}]$$

$$\therefore a_B = 0$$

$$\text{At C ; } \Delta\phi = \frac{2\pi}{4} (1) + \pi = \frac{3\pi}{2}$$

$$\therefore a_C = \sqrt{a^2 + (2a)^2 + 2.a.2.a \cos\left(\frac{3\pi}{2}\right)} = \sqrt{5}a$$

At D : $\Delta\phi =$

$$\Delta\phi = \frac{2\pi}{4} (3) + \pi = \frac{3\pi}{2} + \pi = \frac{5\pi}{2}$$

$$a_D = \sqrt{\left(\frac{a}{2}\right)^2 + (2a)^2 + 2\left(\frac{a}{2}\right)(2a) \cos \frac{5\pi}{2}} = \frac{\sqrt{17}}{2} a$$

6. **Ans (D)**

When potential of inner sphere is V_0 such that $eV_0 = E_2$, then no β particle will be able to touch the outer sphere and will be pushed back to the inner sphere due to the electric field between the spheres. Hence, the final maximum potential of inner sphere is

$$V_0 = \frac{E_2}{e}$$

$$V_0 = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right)$$

Since inner surface of outer sphere and the inner sphere both will have equal and opposite charge

$$\therefore q = \frac{4\pi\epsilon_0 V_0 ab}{b-a} = \frac{4\pi\epsilon_0 ab E_2}{e(b-a)}$$

Charge $-q$ accumulates on the inner wall of the outer sphere due to accumulation of β particles emitted by the inner sphere. There is no supply of charge from the earth.

$$\therefore \text{Charge flow in wire } AB = 0$$

SECTION-I (ii)

7. Ans (C,D)

Consider a Gaussian surface in the shape of a sphere centered at the origin. Radius of the sphere is r .

Flux through the sphere

$$\phi = E \cdot 4\pi r^2 = 4\pi \alpha e^{-\frac{r}{k}}$$

Charge inside the Gaussian sphere

$$q = \epsilon_0 \phi = 4\pi \epsilon_0 \alpha e^{-\frac{r}{k}} \text{ When } r \rightarrow 0 \quad q \rightarrow 4\pi \epsilon_0 \alpha$$

Hence, charge at the origin is $4\pi \epsilon_0 \alpha$ -

When $r \rightarrow \infty \quad q \rightarrow 0$

$\therefore$ Total charge in entire space is zero.

$\therefore$ Charge spread around q_0 is $q = q_0$.

8. Ans (B,C)

Force between capacitor plates is

$$F = \frac{q_0^2}{2\epsilon_0 A}$$

The spring must be compressed by x such that $kx = F$

$$\Rightarrow x = \frac{q_0^2}{2A\epsilon_0 k}$$

$\therefore$ Separation between plates $d = L - x = L$

$-\frac{q_0^2}{2A\epsilon_0 k}$ Capacitance has initial value of

$$C_0 = \frac{\epsilon_0 A}{d} \text{ Energy stored in the capacitor is}$$

$$U_0 = \frac{q_0^2}{2C_0} = \frac{q_0^2 d}{2\epsilon_0 A} = \frac{q_0^2 L}{2\epsilon_0 A} \left(L - \frac{q_0^2}{2A\epsilon_0 k} \right) = \frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{4\epsilon_0^2 A^2 k}$$

Energy stored in the spring is

$$U_s = \frac{1}{2} kx^2 = \frac{q_0^4}{8\epsilon_0^2 A^2 k}$$

The entire energy stored in the capacitor and the spring gets dissipated as heat in the resistor.

$$\therefore \text{Heat} = U_0 + U_s = \frac{q_0^2 L}{2\epsilon_0 A} - \frac{q_0^4}{8\epsilon_0^2 A^2 k}$$

9. Ans (A,B)

Both the gases are heated through isobaric process

$$\Delta Q = n_1 C_{p1} \Delta T + n_2 C_{p2} \Delta T$$

$$= 2 \times \frac{5R}{2} \times \Delta T + 4 \times \frac{7R}{2} \times \Delta T \Rightarrow = 10 k$$

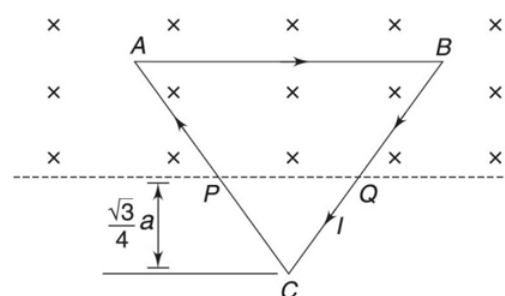
$$(\text{Work done})_B = Q_2 - U_2 =$$

$$n_2 C_{p2} \Delta T - n_2 C_{v2} \Delta T = 40 R \text{ Joules}$$

ΔT will be same not ΔV

$$(\text{Work done})_A = Q_1 - U_1 = 20R \text{ Joules}$$

10. Ans (A,B)



(a) The current in the loop is clockwise.

Magnetic force on the loop will be same as force on a straight wire PQ (current from P to Q)

$$\text{Length PQ} = 2 \frac{\sqrt{3}}{4} a \tan 30 = \frac{a}{2}$$

$$\text{For equilibrium } I \left(\frac{a}{2} \right) B = Mg$$

$$\Rightarrow I = \frac{2Mg}{aB} \dots\dots\dots(i)$$

(b) Assume that the loop is displaced downward by a small distance y .

$$M \frac{d^2 y}{dt^2} = Mg - I \left[2 \left(\frac{\sqrt{3}}{4} a + y \right) \tan 30 \right] B$$

$$\Rightarrow M \frac{d^2 y}{dt^2} = Mg - I \frac{a}{2} B - \frac{2IB}{\sqrt{3}} y$$

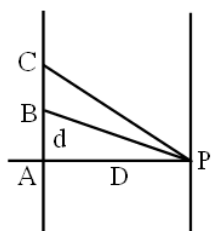
$$\Rightarrow \frac{d^2 y}{dt^2} = - \frac{2IB}{\sqrt{3}M} y \text{ [using (i)]}$$

This is equation of SHM

$$\omega = \sqrt{\frac{2IB}{\sqrt{3}M}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{\sqrt{3}M}{2IB}} = \pi \sqrt{\frac{\sqrt{3}a}{g}} = \text{[using (i)]}$$

11. Ans (A,B,D)



$$BP - AP = \frac{\lambda}{3}$$

$$\sqrt{D^2 + d^2} - D = \frac{\lambda}{3}$$

$$\Rightarrow D \left[1 + \frac{d^2}{D^2} \right]^{\frac{1}{2}} - D = \frac{\lambda}{3}$$

$$\Rightarrow D \left[1 + \frac{d^2}{2D^2} \right] - D = \frac{\lambda}{3}$$

$$[d \ll D \text{ as } BP - AP = \frac{\lambda}{3} \ll D]$$

$$\therefore d = \sqrt{\frac{2D\lambda}{3}} \dots (a)$$

(b) Path difference for two waves

$$\Delta x_{CA} = CP - AP$$

$$= \sqrt{D^2 + (2d)^2} - D$$

$$= D \left(1 + \frac{4d^2}{D^2} \right)^{1/2} - D$$

$$= D \left(1 + \frac{2d^2}{D^2} \right) - D = \frac{2d^2}{D}$$

$$= \frac{4\lambda}{3} \left[\because d = \sqrt{\frac{2D\lambda}{3}} \right]$$

$\therefore$ phase difference

$$\Delta \phi_{CA} = \frac{2\pi}{\lambda} \Delta x_{CA} = \frac{2\pi}{\lambda} \cdot \frac{4\lambda}{3} = \frac{8\pi}{3}$$

(c) Path difference at P for waves from B and A

$$\Delta x_{BA} = \frac{\lambda}{3}$$

$\therefore$ phase difference

$$\Delta \phi_{CA} = \frac{2\pi}{\lambda} \Delta x_{BA} = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{3} = \frac{2\pi}{3}$$

The three waves at P can be represented as

$$y_1 = a \sin \omega t$$

$$y_2 = a \sin \left(\omega t + \frac{2\pi}{3} \right)$$

$$y_3 = a \sin \left(\omega t + \frac{8\pi}{3} \right) = a \sin \left(\omega t + \frac{2\pi}{3} \right)$$

$\therefore$ Resultant wave is

$$y = y_1 + y_2 + y_3$$

$$= a \sin \omega t + 2a \sin \left(\omega t + \frac{2\pi}{3} \right)$$

$$= A \sin (\omega t + \delta)$$

Where

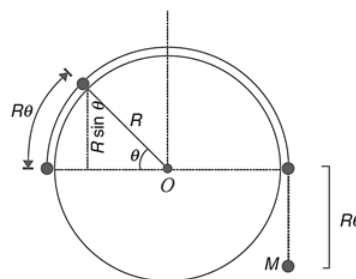
$$A = \sqrt{a^2 + (2a)^2 + 2 \cdot a \cdot 2a \cdot \cos \frac{2\pi}{3}} = \sqrt{3}a.$$

Intensity is proportional to square of amplitude

$$\therefore I = 3I_0$$

12. Ans (A,B,D)

(a) When M descends through $R\theta$, radius vector of m rotates through θ as shown in figure. The normal contact force does not perform any work. Only other force is gravity (which is conservative).



$\therefore$ loss in PE = gain in KE

Loss in PE of M - gain in PE of m =

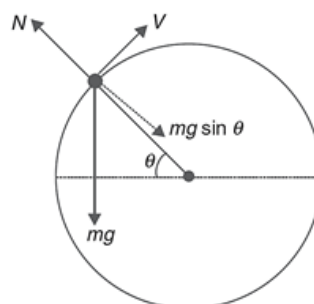
$$\frac{1}{2} M v^2 + \frac{1}{2} m v^2$$

(Since speed of both M and m will be same.)

$$\therefore MgR\theta - mg(R \sin \theta) = \frac{1}{2} (M + m) v^2 \therefore v =$$

$$\sqrt{2gR \frac{(M\theta - m \sin \theta)}{M + m}}$$

(b) Consider the motion of m at this instant.



$$\begin{aligned}\frac{mv^2}{R} &= mg \sin \theta - N \Rightarrow N = mg \sin \theta - \frac{mv^2}{R} \\ \frac{mv^2}{R} &= mg \sin \theta - \frac{2mgR(M\theta - m \sin \theta)}{(M+m)R} \\ &= mg \left[\frac{(M+m) \sin \theta - 2M\theta + 2m \sin \theta}{M+m} \right] \\ &= \frac{mg}{M+m} [(M+3m)\sin \theta - 2M\theta]\end{aligned}$$

$$\theta \rightarrow 0$$

(c) Just after release $\sin \theta = \theta \therefore N = \frac{mg}{M+m}$

$$[(M+3m)\theta - 2M\theta] = \frac{mg}{M+m} [(3m-M)\theta]$$

For contact to be maintained $N > 0$

$$\Rightarrow 3m > M$$

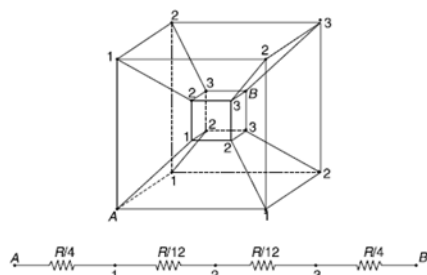
SECTION-II

1. Ans (20.00)

Due to symmetry

Point marked 1 are equipotential points.

Similarly all 2s are at same potential and all 3s are also at same potential



2. Ans (0.4)

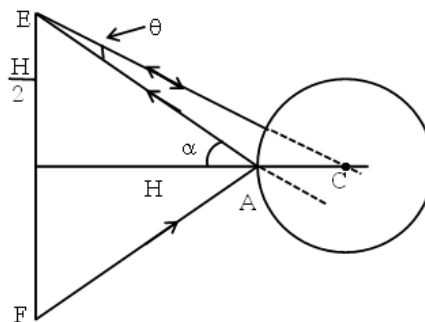
A light ray FA will get reflected and reach the eye of the man travelling along AE. It means image of F will be seen on line EA extended. Similarly a ray from E heading towards the centre of the ball will get reflected back along the same line (CE) and therefore image of E will lie somewhere on the line EC.

$\therefore$ Angle subtended by the image at the eye is angle

$$\angle AEC = \theta \text{ (say)}$$

In $\triangle AEC$: $\angle C = \alpha$ [$\because AC = r$ is very small]

$$AE = \sqrt{H^2 + \frac{H^2}{4}} = \frac{\sqrt{5}}{2}H$$



Applying sine rule in $\triangle AEC$

$$\frac{\sin \theta}{AC} = \frac{\sin \alpha}{AE}$$

$$\Rightarrow \sin \theta = \frac{r}{AE} \cdot \frac{H/2}{AE} = \frac{rH}{2(AE)^2}$$

$$\Rightarrow \sin \theta = \frac{rH}{2 \cdot \frac{5}{4}H^2} = \frac{2r}{5H}$$

Since θ is small $\sin \theta = \theta$

$$\therefore \theta \simeq \frac{2r}{5H}$$

3. Ans (4.00)

Point Q is the centre of a cube of side length L,

one of the faces of which is the plate. One sixth of the total radiation strikes the plate.

Point P is centre of a cube of side length 2 L

and the given plate is

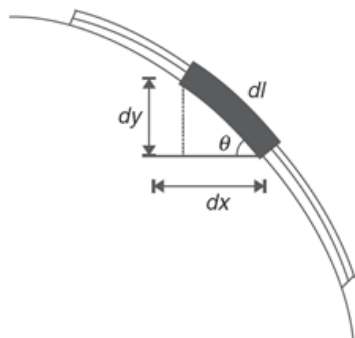
$\frac{1}{4}$ one face of cube. Hence radiation striking

the plate is $\frac{1}{24}$ of the total radiation emitted.

$$\therefore F = 4 F_0$$

4. Ans (0.4)

Consider a small segment of length dl on the rope.



The tangential component of weight

$$= \lambda dl g \sin \theta = \lambda g (dl \sin \theta) = \lambda g dy$$

For complete rope, sum of tangential component of weight is

$$W_t = \lambda g \int_0^H dy = \lambda g H$$

Maximum friction force on the segment

$$= \mu (\lambda dl g \cos \theta) = \mu \lambda g (dl \cos \theta) = \mu \lambda g dx$$

Sum of friction on the complete rope

$$f_{\max} = \mu \lambda g \int_0^x dx = \mu \lambda g x$$

Rope will be in equilibrium if $f_{\max} \geq W_t$

$$\mu \lambda g x \geq \lambda g H \quad \mu \geq \frac{H}{x}$$

5. Ans (440.00)

Average of rectifier output is $\frac{2\varepsilon_0}{\pi} = 8.98$

$$\therefore \varepsilon_0 = \frac{8.98 \times 3.14}{2} = 14.1$$

$\therefore$ rms value of transformer output (i.e., same as rectifier input)

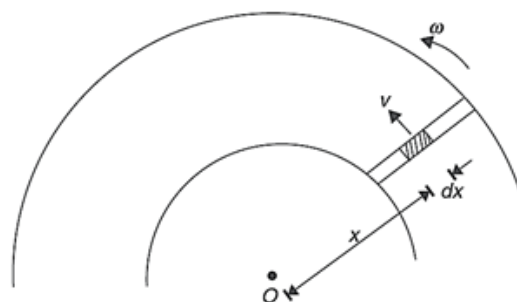
$$= \frac{\varepsilon_0}{\sqrt{2}} = \frac{14.1}{1.41} \simeq 10 \text{ volt}$$

$$\frac{\varepsilon_{\text{input}}}{\varepsilon_{\text{output}}} = \frac{N_p}{N_s}$$

$$\frac{220}{10} \times 20 = N_p$$

$$\therefore N_p = 440$$

6. Ans (3.00)



Consider an element of length dx on the blade.

Speed of this element is $v = \omega x$

Viscous force on this element is

$$dF = \eta A \left(\frac{v}{t} \right)$$

$$\left[\frac{v}{t} = \text{velocity gradient} \right] \frac{v}{t}$$

$$= \eta b dx \left(\frac{\omega x}{t} \right) = \eta \frac{b\omega}{t} \cdot x dx$$

Torque due to this force

$$d\tau = x dF = \frac{\eta b\omega}{t} x^2 dx$$

$\therefore$ Net viscous torque on the blade is

$$\tau = \frac{\eta b\omega}{t} \int_R^{R+L} x^2 dx \Rightarrow \tau = \frac{\eta b\omega}{t} \frac{1}{3} [x^3]_R^{R+L}$$

$$= \frac{\eta b\omega}{4t} [(R+L)^3 - R^3]$$

$$= \frac{\eta b\omega R^3}{3t} \left[\left(1 + \frac{L}{R} \right)^3 - 1 \right]$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (D)

($\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$) Mohr salt

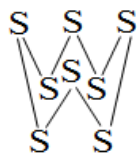
$$\lambda_m^\circ(\text{salt}) = \lambda_m^\circ(\text{Fe}^{2+}) + 2\lambda_m^\circ(\text{NH}_4^+)$$

$$+ 2\lambda_m^\circ(\text{SO}_4^{2-}) = a + 2b + 2c$$

$$\lambda_{\text{eq}}^\circ(\text{salt}) = \frac{\lambda_m^\circ(\text{salt})}{\nu f} = \frac{a + 2b + 2c}{4}$$

2. Ans (B)

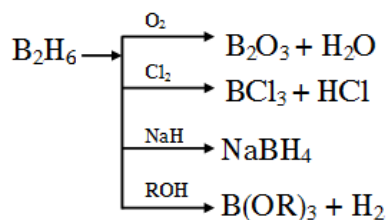
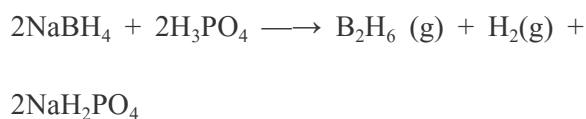
In crown conformation of S₈ molecule, on connecting S atoms it look like square antiprismane geometry



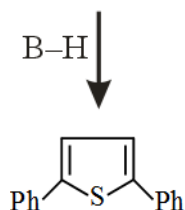
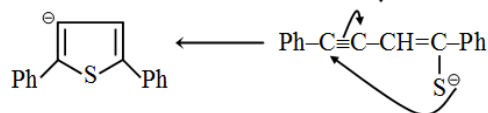
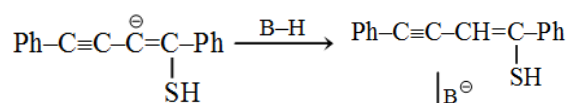
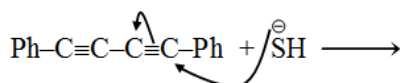
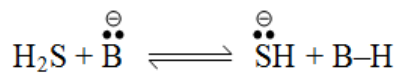
3. Ans (C)

Factual

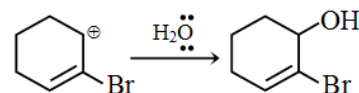
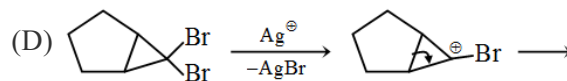
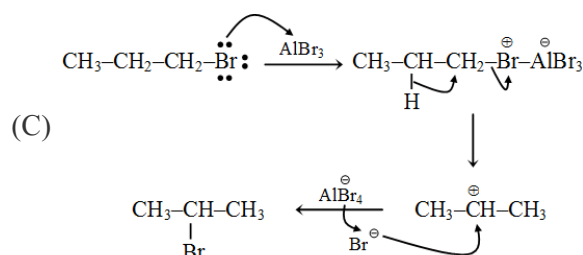
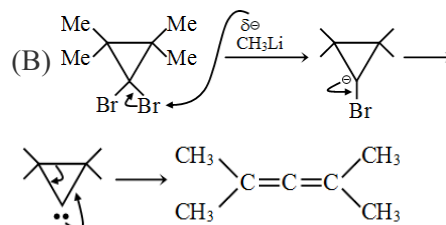
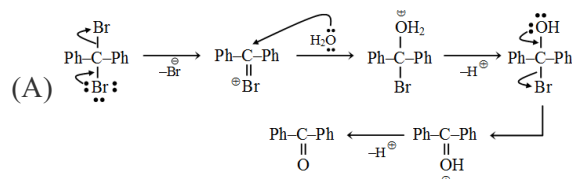
4. Ans (D)



5. Ans (D)



6. Ans (D)



SECTION-I (ii)

7. Ans (A,B,C)

(A) K[PtCl₃(C₂H₄)] has EAN of 84

(B) With respect to any two group, there are two cis and one trans

(C) M(AB)₂ (tetrahedral) has no POS or COS

(D) [Co(EDTA)]⁻ can show OI but not GI.

8. Ans (B)

(A) P_1 & P_2 cannot be compared as both are p_{H_2O}

in different-different equilibria.

(B) Na_2SO_4 will start converting to $Na_2SO_4 \cdot 10H_2O$

when p_{H_2O} exceeds P_2 . But if $P_1 < P_2$, it is not

possible.

(C) If $P_1 > P_2$, 2nd equilibrium will shift backward.

But nothing has been said about amount of

Na_2SO_4 . So all $H_2O(l)$ may or may not dry up

eventually.

(D) If amount of $H_2O(l)$ is very less such that

$p_{H_2O(g)} < P_2$, then neither 2nd equilibrium will shift

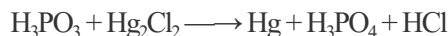
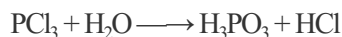
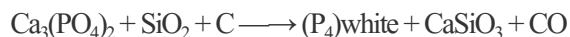
backward nor 1st equilibrium will be attained. So,

$(p_{H_2O(g)}) < P_2 < P_1$.

9. Ans (A,B,C,D)

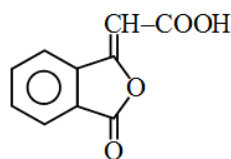
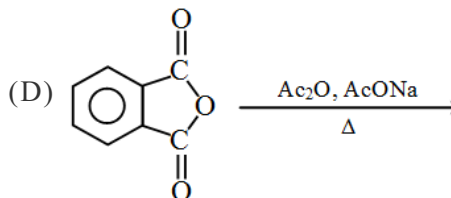
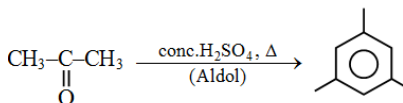
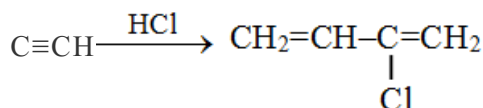
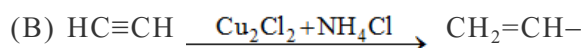
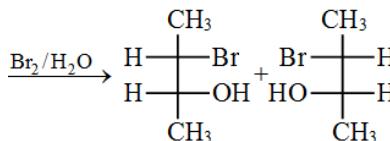
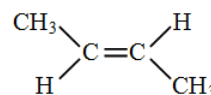
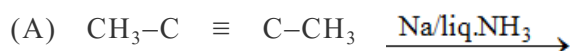
bond order in graphite = $1 + \frac{1}{3} = \frac{4}{3}$

10. Ans (A,B,C,D)



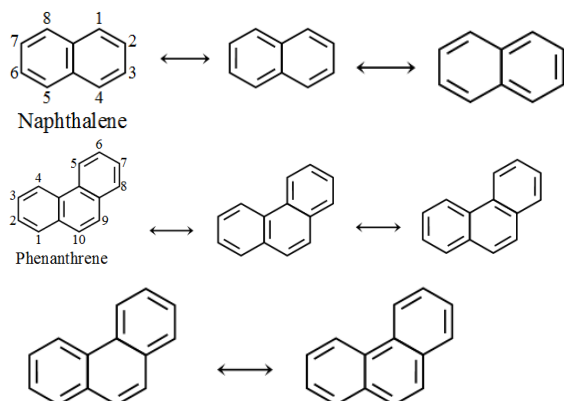
S_2Cl_2 has open book structure like H_2O_2

11. Ans (B,C,D)



(Similar to Perkin reaction.)

12. Ans (A,B,C,D)



Molecular orbital calculation of naphthalene show bond orders of 1.724 and 1.603 respectively (compare benzene, 1.667). In agreement with these predictions the 1, 2 and 2, 3 bond distance are 1.36 and 1.415 Å, respectively. This non equivalency of bonds, called partial bond fixation, is found in nearly all fused aromatic system. In phenanthrene where the 9, 10 bond is a single bond in only one of five forms, bond fixation becomes extreme and this bond is readily attacked by many reagents.

SECTION-II

1. Ans (1.62)

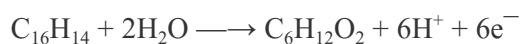
$$1.2 \times 10^{-2} = \frac{7.2 \times 10^{-2} \alpha^2}{1 - \alpha}$$

$$\Rightarrow 6\alpha^2 + \alpha - 1 = 0 \quad \alpha = \frac{1}{3}$$

$$\text{So, pH} = -\log \frac{0.072}{3} = 1.62$$

(solution will be of a weak monoprotic acid)

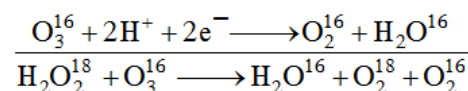
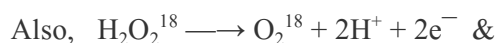
2. Ans (18.00)



∴ Formation of 1 mole P from A requires 6 F charge. So, x = 6

Out of the 5 statement, (i), (ii) & (iv) are correct. So y = 3

∴ Reported answer = 6 × 3 = 18



Since O₃ is a better OA than H₂O₂.

3. Ans (1.50)

$$\text{Moles of NaOH} = 3 \times 1 = 3$$

$$\therefore \text{moles of H}_2\text{SO}_4 = 1.5$$

Let molarity of H₂SO₄ is 'x' then

$$1.5 = xv \quad \dots\dots (1)$$

For diluted solution

$$\text{Moles of NaOH} = 3 \times 0.2 = 0.6$$

$$\therefore \text{moles of H}_2\text{SO}_4 = 0.3$$

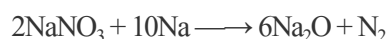
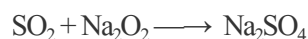
$$xv = M_2 \cdot 5 \quad \therefore M_2 = \frac{xv}{5}$$

$$0.3 = \frac{xv}{5} \cdot v \quad \text{or} \quad 1.5 = xv^2 \quad \dots\dots (2)$$

$$\text{From 1 \& 2} \quad v = 1$$

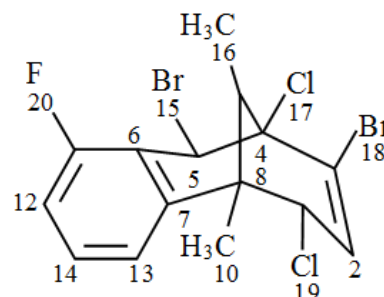
$$\text{Conc. of H}_2\text{SO}_4 = 1.5 \text{ M}$$

4. Ans (0.25)



5. Ans (6.00)

Let us number the atoms for convenience of discussion (this may not be as per IUPAC nomenclature)

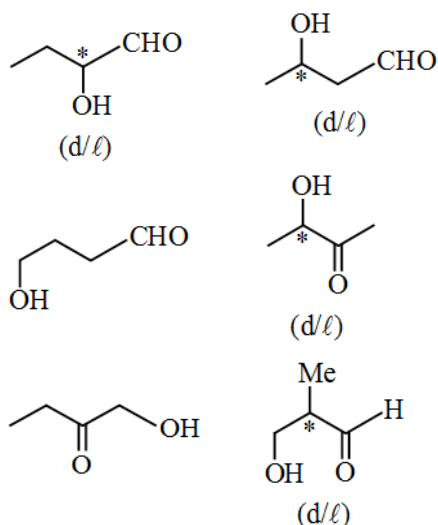


(a) all halides except -F can form Grignard reagent on addition of Mg. Reaction of vinylic halogen is slow, but possible. Hence (P) = 4.

(b) can produce a precipitate with aq. AgNO₃ is (Q) = 2 (Br-15 and Cl-19)

(c) can give a substitution reaction with KCN is (R) = 0 (S_N1 predominates at Br-15 and Cl-19, others inert)

6. Ans (10.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$I = \frac{1}{4} \int \frac{4x^2 dx}{(x^4 - 1)\sqrt{x^4 + 1}}$$

$$\Rightarrow I = \frac{1}{4} \int \frac{((x^2 + 1)^2 - (x^2 - 1)^2)}{(x^2 - 1)(x^2 + 1)\sqrt{x^4 + 1}} dx$$

$$I = \frac{1}{4} \int \frac{(x^2 + 1)}{(x^2 - 1)\sqrt{x^4 + 1}} dx - \frac{1}{4} \int \frac{(x^2 - 1)}{(x^2 + 1)\sqrt{x^4 + 1}} dx$$

$$\text{Let } I_1 = \frac{1}{4} \int \frac{(x^2 + 1)}{(x^2 - 1)\sqrt{x^4 + 1}} dx,$$

$$I_2 = \frac{1}{4} \int \frac{(x^2 - 1)}{(x^2 + 1)\sqrt{x^4 + 1}} dx$$

Now in I_1 put $x - \frac{1}{x} = t$ and in I_2 put $x + \frac{1}{x} = k$
then

$$I = \frac{1}{4} \int \frac{dt}{t\sqrt{t^2 + 2}} - \frac{1}{4} \int \frac{dk}{k\sqrt{k^2 - 2}}$$

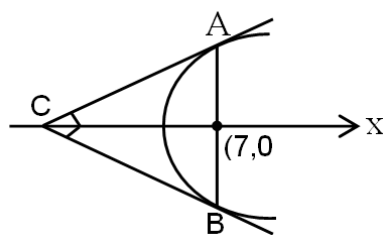
Now put $t^2 + 2 = m^2$ and $k^2 - 2 = n^2$

$$I = \frac{1}{4} \int \frac{dm}{(m^2 - 2)} - \frac{1}{4} \int \frac{dn}{(n^2 + 2)}$$

$$= \frac{1}{4} \cdot \frac{1}{2\sqrt{2}} \ln \left(\frac{m - \sqrt{2}}{m + \sqrt{2}} \right) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \frac{n}{\sqrt{2}}$$

$$= \frac{1}{4\sqrt{2}} \left[\log \left(\frac{\sqrt{x^4 + 1} - x\sqrt{2}}{x^2 - 1} \right) - \tan^{-1} \left(\frac{\sqrt{x^4 + 1}}{x\sqrt{2}} \right) \right] + c$$

2. Ans (A)



$x = 7 + ty$ is a focal chord of $y^2 = 28x$

Tangents at the ends of focal chord are always $\perp$

$\therefore$ Circumcentre is the mid point of AB

Let mid point is (h, k)

$$\Rightarrow T = S_1$$

$$ky - 14(x + h) = k^2 - 28h$$

compare with $x = 7 + ty$

$$14 = \frac{k}{t} = \frac{28h - 14h - k^2}{7} \Rightarrow 98 = 28h - 14h - k^2$$

$$y^2 = 14(x - 7)$$

3. Ans (D)

$$\sum_{r=1}^{20} {}^{20}C_r \left(\sum_{p=0}^{(r-1)} {}^r C_p \cdot 5^p \right)$$

$$\sum_{r=1}^{20} {}^{20}C_r (6^r - 5^r)$$

$$\sum_{r=1}^{20} {}^{20}C_r \cdot 6^r - \sum_{r=1}^{20} {}^{20}C_r 5^r$$

$$7^{20} - 6^{20}$$

4. Ans (A)

'n' should be odd for $f(x)$ to be invertible.

Here maxima and minima cannot occur. So

roots of $f'(x) = 0$ is also the roots of $f''(x) = 0$

between any two roots of $f'(x) = 0$ there exist atleast one roots of $f''(x) = 0$ (Rolle's theorem)

$\Rightarrow$ If $f'(x) = 0$ has k distinct root then all are repeated '2m' times ($m \in \mathbb{N}$).

$\Rightarrow$ If $f'(x) = 0$ has $\frac{n-1}{2}$ distinct real root then

$$\text{equation } f''(x) = 0 \text{ has } \frac{(n-1)}{2} + \frac{(n-1)}{2} - 1$$

$= (n-2)$ distinct roots (Because Rolle's says $f''(x) = 0$ has atleast one root between each consecutive root of $f'(x) = 0$)

5. **Ans (D)**

$${}^{16}C_0 - ({}^{16}C_0 + {}^{16}C_1) - ({}^{16}C_0 + {}^{16}C_1 + {}^{16}C_2) + ({}^{16}C_0 + {}^{16}C_1 + {}^{16}C_2 + {}^{16}C_3) - \dots + ({}^{16}C_0 + {}^{16}C_1 + \dots + {}^{16}C_{n-1}) = -{}^{16}C_1 + {}^{16}C_3 - {}^{16}C_5 + \dots + {}^{16}C_{15} = 0$$

6. **Ans (C)**

$$x_1 + x_2 = -a, x_1 x_2 = b$$

given condition

$$16x_1 x_2 = 2(x_1 + x_2) - 1$$

$$16b = -2a - 1$$

$$16b + 2a = -1$$

given $f(x)$ is differentiable

$$a^2 = a\alpha + b \text{ \& } 2\alpha = a$$

$$\Rightarrow \frac{a^2}{4} = \frac{a^2}{2} + b \Rightarrow b = -\frac{a^2}{4}$$

$$a = \frac{1 \pm \sqrt{5}}{4} \Rightarrow \frac{1 + \sqrt{5}}{4}$$

$$a = \cos 36^\circ$$

SECTION-I (ii)

7. **Ans (A,C,D)**

$\therefore f(x)$ and $g(x)$ are polynomials of degree 3

then suppose $g(x) = \lambda(x - \alpha)^2(x - \beta)$ ($\alpha \neq \beta$)

and for existence of limit $f(x) = 0, f'(\alpha) = 0, f''(\alpha) = 0$

$$\therefore f(x) = \mu(x - \alpha)^3$$

$$\therefore f(x).g(x) = \lambda.\mu.(x - \alpha)^5(x - \beta)$$

$$\therefore f(x).g(x) = \lambda.\mu.(x - \alpha)^5(x - \beta)$$

$$\text{Also, } h(x) = f(x).g'(x) + g(x).f''(x) =$$

$$\frac{d}{dx}(f(x).g(x))$$

$$= \frac{d}{dx}(\lambda\mu(x - \alpha)^5(x - \beta))$$

$$= \lambda\mu[(x - \alpha)^5 \cdot 1 + 5(x - \beta).(x - \alpha)^4]$$

$$= \lambda\mu(x - \alpha)^4[(x - \alpha) + 5x - 5\beta]$$

$$h(x) = \lambda\mu(x - \alpha)^4[6x - \alpha - 5\beta]$$

$\therefore h(x) = 0$ has two distinct roots $x = \alpha$ and

$$x = \frac{\alpha + 5\beta}{1 + 5}$$

which lies between α and β . Clearly $h'''(\alpha) = 0$.

8. **Ans (A,B,D)**

$f(x)$ depends on $\sin^{-1}x$ & $\cos x$, then both are continuous function

$$f(x) = \sin^{-1} \left(\frac{2}{(1 - \cos x) + (1 + \cos x)} \right)$$

$$\sin^{-1}(1) = \frac{\pi}{2}$$

$f(x)$ is a constant function and it is many one

$$f(|x|) = \frac{\pi}{2}$$

9. **Ans (A,B,C,D)**

$$\left(\frac{x^2 + 2}{x^2 + 1} \right)^2 - (a + 4) \left(\frac{x^2 + 2}{x^2 + 1} \right) + (a + 3) = 0$$

$$\text{put } \frac{x^2 + 2}{x^2 + 1} = t$$

$$\Rightarrow t = 1 + \frac{1}{x^2 + 1} \Rightarrow t \in (1, 2]$$

$$\Rightarrow t^2 - (a + 4)t + (a + 3) = 0$$

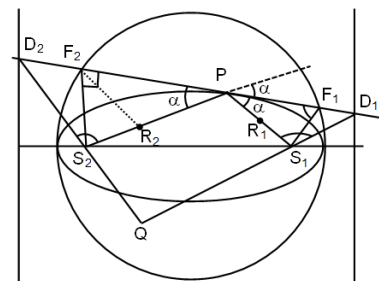
$$\Rightarrow (t - 1)(t - (a + 3)) = 0$$

$$\Rightarrow t = 1 \quad t = a + 3$$

$$\text{so } +1 < a + 3 \leq 2$$

$$-2 < a \leq -1$$

10. **Ans (A,C)**



$$\text{Let } \angle D_2 P S_2 = \angle D_1 P S_1 = \alpha$$

$\{\therefore \text{tangent bisects angle } S_1 P S_2 \text{ externally}\}$

$$\text{Then } P F_2 = P S_2 \cos \alpha, P F_1 = P S_1 \cos \alpha$$

$$\Rightarrow F_1 F_2 = P F_1 + P F_2 = (P S_1 + P S_2) \cos \alpha = a$$

$$\Rightarrow 2a \cos \alpha = a \Rightarrow \cos \alpha = \frac{1}{2} \Rightarrow \alpha = 60^\circ$$

$$\angle P D_1 S_1 = 90^\circ - 60^\circ = 30^\circ$$

$$\Rightarrow \Delta D_1 D_2 Q \text{ is isosceles } \Rightarrow D_1 Q = D_2 Q$$

11. **Ans (A,B,C)**

Length of sides of cuboid are either 1, 2, 6 or 1, 4, 3 or 2, 2, 3

$\Rightarrow$ Length of body diagonal of cuboid can be

$$\sqrt{41}, \sqrt{26}, \sqrt{17}$$

12. Ans (A,B,C)

$$A^2 + I = 2B$$

$$A^2 - 2A + I = 2(B - A)$$

$$(A - I)^2 = 2(B - A)$$

$$|A - I|^2 = 16 |B - A| \neq 0$$

$A - I$ is invertible matrix

Similarly $B - I$ is invertible matrix

$$\text{Now } A^2 - I = 2(B - I)$$

$$\Rightarrow |A - I| |A + I| = 16 |B - I|$$

$$\Rightarrow |A + I| \neq 0$$

Similarly $|B + I| \neq 0$

$$\text{Now } B = \frac{(A^2 + I_4)}{2}$$

$$\Rightarrow A^4 + 2A^2 + I_4 - 8 + 4I_4 = O_4$$

$$\Rightarrow A^4 + 2A^2 - 8 + 5I_4 = O_4$$

$$\Rightarrow (A - I)^2 (A^2 + 2A + 5I) = O$$

$$\Rightarrow A^2 + 2A + 5I_4 = O$$

$$\Rightarrow 2B - I_4 + 2A + 5I_4 = O$$

$$\Rightarrow A + B = -2I_4$$

$$\Rightarrow |A + B| = 16$$

$$\text{Also } A^2 + 2A + I_4 = -4I_4$$

$$\Rightarrow |A + I|^2 = 4^4 \Rightarrow |A + I| = 256$$

SECTION-II

1. Ans (0.00)

$$\text{Given } f(x) + f(-x) = 5$$

$$\text{Let } f(x) = a$$

$$\Rightarrow f(-x) = 5 - a$$

$$\Rightarrow f^{-1}(a) + f^{-1}(5 - a) = 0$$

$$\text{Let } I = \int_{1-x}^{4+x} f^{-1}(t) dt \dots (1)$$

using king property $t \rightarrow 5 - t$

$$I = \int_{1-x}^{4+x} f^{-1}(5 - t) dt \dots (2)$$

adding

$$2I = \int_{1-x}^{4+x} (f^{-1}(t) + f^{-1}(5 - t)) dt = \int_{1-x}^{4+x} 0 dt$$

$$\Rightarrow I = 0$$

2. Ans (0.50)

$$|\sin x| + |\cos x| = \sqrt{\sin^2 x + \cos^2 x + 2|\sin x||\cos x|}$$

$$= \sqrt{1 + |\sin 2x|} \geq 1$$

$$\therefore y = [|\sin x| + |\cos x|] = 1$$

Now, point of intersection of $y = 1$ & $xy = 2$ is $(2, 1)$

$$y = \frac{2}{x}$$

$$\frac{dy}{dx} = -\frac{2}{x^2} \therefore \left. \frac{dy}{dx} \right|_{x=2} = -\frac{1}{2}$$

$$m_2 = 0 \quad \{\text{slope of } y = 1\}$$

$$\therefore \text{Angle of intersection is } \theta = \tan^{-1} \frac{1}{2}$$

3. Ans (0.54)

$$\tan \alpha = \frac{BC}{2AC} \text{ and } \tan \beta = \frac{2BC}{AC},$$

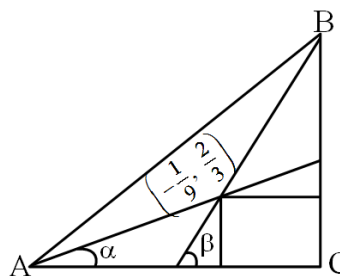
thus if m_1 and m_2 be the slopes of medians through A and B

$$\text{then } \frac{m}{3} = 4 \text{ or } \frac{1}{4}$$

$$\Rightarrow m = 12, \frac{3}{4} \text{ (Rejected)}$$

$$\frac{BC}{AC} = 6$$

$$\Rightarrow AC = 2 \text{ \& } BC = 12$$



centroid is point of intersection of $y = 3x + 1$ &

$$y = 12x + 2 \text{ which is } \left(-\frac{1}{9}, \frac{2}{3}\right)$$

$$x \text{ coordinates of C is } -\frac{1}{9} + \frac{2}{3} = \frac{5}{9}$$

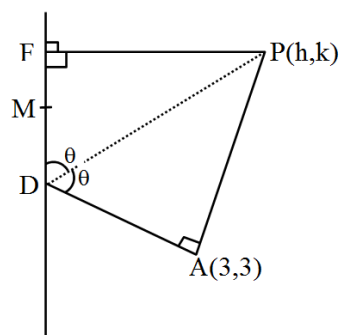
$$y \text{ coordinates of C is } \frac{2}{3} - 4 = -\frac{10}{3}$$

4. Ans (1.08)

Middle term in each row is square of the row number

$$\text{Sum of terms in } n^{\text{th}} \text{ row} = n^3$$

5. Ans (23.00)



In $\triangle FDP$ & $\triangle ADP$

$\angle F = \angle A = 90^\circ$, $\angle FDP = \angle ADP = \theta$ & PD is common

$\Rightarrow \triangle FDP$ & $\triangle ADP$ are congruent $\Rightarrow PF = PA$

$$\Rightarrow (h-3)^2 + (k-3)^2 = \left(\frac{h+k-2}{\sqrt{2}} \right)^2$$

$$\Rightarrow x^2 + y^2 - 2xy - 8x - 8y + 32 = 0$$

6. Ans (1.00)

$$h(\omega) = \omega f(1) + \omega^2 g(1) = 0 \dots (i)$$

$$h(\omega^2) = \omega^2 f(1) + \omega g(1) = 0 \dots (ii)$$

from (1) & (2) we get

$$f(1) = 0, g(1) = 0$$

$$\text{Now } h(1) = f(1) + g(1) = 0$$

$$\Rightarrow h(-1) = -f(1) + g(1) = 0$$

$$\Rightarrow h(-\omega) = 0$$

$$\Rightarrow h(-\omega^2) = 0$$

$$\Rightarrow h(0) = 0$$

$\Rightarrow$ equation $h(x) = 0$ has

at least 7 root $1, -1, \omega, -\omega, \omega^2, -\omega^2, 0$

$\Rightarrow$ minimum degree of $h(x)$ is 7

If degree of $h(x)$ is 7

then $h(x)$ is $(x+1)(x-1)(x-\omega)(x+\omega)(x+\omega^2)(x-\omega^2)(x)$

$$\Rightarrow h(x) = x^7 - x \Rightarrow h(2) = 126$$

If $f(x)$ and $g(x)$ are polynomial then let $f(x) = ax + b$ and $g(x) = c$

$$\Rightarrow h(x) = x(ax^6 + b) + x^2(c)$$

$$\Rightarrow x^7 - x = ax^7 + bx + cx^2$$

$$\Rightarrow c = 0, a = 1, b = -1$$

$$\Rightarrow f(x) = x - 1 \text{ and } f(2) = 1$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

Full Syllabus

14-08-2021

JEE(Main + Advanced) : Leader & Enthusiast Course (PHASE : SCORE-3.0)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,B,C,D	B	A	A,B,C	A,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.66 to 2.67	25.00 to 25.20	12.00	2.66 to 2.67	0.50	3.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	8	4	9	3	3	1

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C,D	A,C,D	C,D	C	A,B,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	96.00	4.75	16.00	0.11	9.00	10.00
SECTION-III	Q.	1	2	3	4	5	6
	A.	6	4	7	1	6	8

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	B,C	A,B,D	A,C,D	A,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	1275.00	960.00	12.00	2.00	324.00	12.80
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	4	4	1	3	8

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A,B,C,D)

The given system is equivalent to the one shown in the Figure. The distance between the plates of C_2 is $\frac{5}{2}t$.

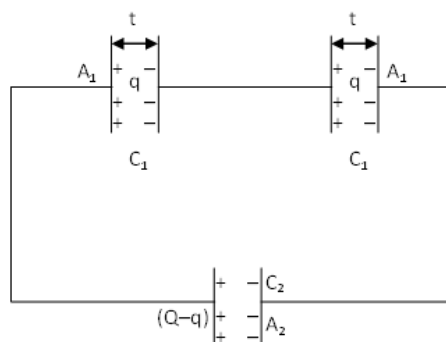


Plate area of capacitors represented by C_1 and C_2 are $A_1 = xL$ and $A_2 = L(L-x)$

$$C_1 = \frac{\epsilon_0 A_1}{t} = \frac{\epsilon_0 Lx}{t} ;$$

$$C_2 = \frac{\epsilon_0 A_2}{\frac{5t}{2}} = \frac{2\epsilon_0 L(L-x)}{5t}$$

Let charge on capacitors having capacitance C_1 be q and that on capacitors of capacitance C_2 be $Q-q$. The equivalent of two C_1 in series is $\frac{C_1}{2}$.

$\therefore$ Potential difference across $\frac{C_1}{2}$ = p.d across C_2 .

$$\therefore \frac{q}{C_{1/2}} = \frac{Q-q}{C_2}; \frac{2q}{\frac{\epsilon_0 Lx}{t}} = \frac{(Q-q)}{\frac{2\epsilon_0 L(L-x)}{5t}}$$

$$\Rightarrow \frac{2q}{x} = \frac{5(Q-q)}{2(L-x)} \Rightarrow 4q(L-x) = 5(Q-q)x$$

$$\therefore q[4L-4x+5x] = 5Qx \therefore q = \frac{5Qx}{4L+x}$$

$$\text{And } Q-q = \frac{4Q(L-x)}{4L+x}$$

$$(a) \sigma_1 = \frac{Q-q}{L(L-x)} = \frac{4Q}{L(4L+x)}; \sigma_2 = \frac{q}{Lx} = \frac{5Q}{L(4L+x)}$$

$$(b) \text{ Required p.d.} = \text{p.d. across } C_1 = \frac{q}{C_1}$$

$$= \frac{5Qx}{4L+x} \cdot \frac{t}{\epsilon_0 Lx} = \frac{5Qt}{\epsilon_0 L(4L+x)}$$

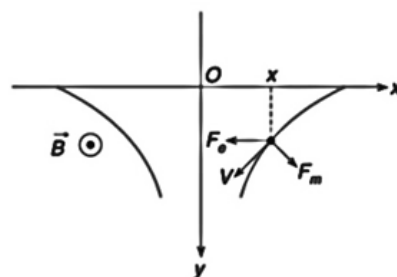
$$(c) E = \frac{\sigma_1}{\epsilon_0} = \frac{4Q}{\epsilon_0 L(4L+x)}$$

$$(d) C = C_2 + \frac{C_1}{2} = \frac{2\epsilon_0 L(L-x)}{5t} + \frac{\epsilon_0 Lx}{2t}$$

$$= \frac{\epsilon_0 L(4L+x)}{10t}$$

2. Ans (B)

(a) The two particles follow symmetrical curved paths. As the electrostatic attraction speeds them, the magnetic force curves the path. Magnetic force does not perform any work, therefore any work, therefore conservation of energy gives



$$\frac{1}{2}mV^2 \times 2 - k\frac{q^2}{(2x)} = -1\frac{q^2}{(2a)}$$

$$\Rightarrow 2mV^2 = kq^2 \left(-\frac{1}{a} + \frac{1}{x} \right)$$

$$\Rightarrow V = \sqrt{\frac{kq^2}{2m} \left(\frac{1}{x} - \frac{1}{a} \right)}$$

(b) Force on $-q$ is

$$\vec{F} = -F_e \hat{i} + (-q)(V_x \hat{i} + V_y \hat{j}) \times (-B\hat{k})$$

$$= (-F_e + qV_y B) \hat{i} - qB V_x \hat{j}$$

$$\therefore m \frac{dV_y}{dt} = -qB V_x \Rightarrow m \frac{dV_y}{dt} = -qB \frac{dx}{dt}$$

$$\Rightarrow \int_0^{V_y} dV_y = -\frac{qB}{m} \int_a^x dx \Rightarrow V_y = \frac{qB}{m}(a-x)$$

3. Ans (A)

In steady state current density will be uniform.

$$J = \rho(x)v(x)$$

$$\text{By Gauss' Law } (E + dE)A - EA = \frac{\rho A dx}{\epsilon_0}$$

$$\frac{dE}{dx} = \frac{\rho(x)}{\epsilon_0} = \frac{J}{\epsilon_0 v(x)}$$

$v(x)$ increases with 'x'

$\therefore \rho(x)$ decreases with 'x'.

4. **Ans (A,B,C)**

(a) velocity of block = v_0

Angular velocity of the pulley $\omega = \frac{v_0}{a}$

Emf developed across the circumference and centre

$$\varepsilon = \frac{1}{2} B \omega a^2 \quad \text{Current in the resistance, } i = \frac{B \omega a^2}{2R}$$

$$\text{Power dissipated in } R \text{ is } P_R = i^2 R = \frac{B^2 \omega^2 a^4}{4R}$$

Mechanical power delivered to pulley = rate of change of its kinetic energy

$$P_{\text{mech}} = \frac{d}{dt} \left(\frac{1}{2} I \omega^2 \right) = I \omega \frac{d\omega}{dt} \quad \dots(i)$$

$P_R + P_{\text{mech}}$ = power delivered by string tension to the pulley = $T \cdot a \cdot \omega$

$$= \left[Mg - M \frac{dv}{dt} \right] a \omega \quad \left[\because T = Mg - M \frac{dv}{dt} \right]$$

$$\therefore \frac{B^2 \omega^2 a^4}{4R} + I \omega \frac{d\omega}{dt} = Mga\omega - Ma\omega \frac{d(\omega a)}{dt}$$

$$[\because v = \omega a]$$

$$\Rightarrow [I + M \cdot a^2] \frac{d\omega}{dt} = Mga - \frac{B^2 a^2 \omega}{4R}$$

Acceleration of mass M is $\left(a \frac{d\omega}{dt} \right)$

$$\therefore a_M = \frac{Mga^2 - \frac{B^2 a^5 \omega}{4R}}{I + Ma^2}$$

$$\text{When } \omega = \frac{v_0}{a}$$

$$a_M = \frac{Mga^2 - \frac{B^2 a^4 v_0}{4R}}{I + Ma^2} \quad \dots(i)$$

(b) when the block acquires terminal speed $a_M = 0$

$$\Rightarrow Mga^2 = \frac{B^2 a^4 v_T}{4R} \Rightarrow v_T = \frac{4MgR}{B^2 a^2}$$

$$(c) \text{ When } \omega a = \frac{v_T}{2} \quad \omega = \frac{2MgR}{B^2 a^3}$$

$$\frac{d\omega}{dt} = \frac{a_M}{a} = \frac{Mga - \frac{B^2 a^3}{4R} \cdot \frac{2MgR}{B^2 a^2}}{I + Ma^2}$$

$$= \frac{Mga}{2(I + Ma^2)}$$

$$\text{From (i) } P_{\text{mech}} = I \omega \frac{d\omega}{dt}$$

$$= \frac{2MgR}{B^2 a^3} \cdot \frac{Mga}{2(I + Ma^2)} = \frac{I \cdot M^2 g^2 R}{B^2 a^2 (I + Ma^2)}$$

6. **Ans (A,B,C)**

(a) Consider only the stick at an angle θ

Torque about O, $\tau = Mg$

$$\frac{L}{2} \cos \theta$$

Angular acceleration

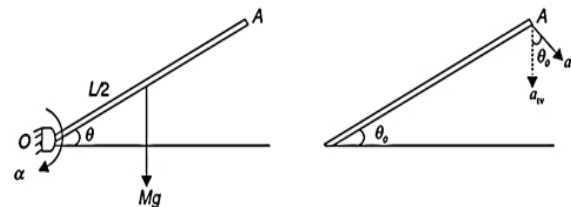
$$\alpha = \frac{\tau}{I} = \frac{Mg \frac{L}{2} \cos \theta}{\frac{1}{3} ML^2} = \frac{3}{2} \frac{g}{L} \cos \theta$$

The tangential acceleration of end A is :

$$a_t = L\alpha = \frac{3}{2} g \cos \theta$$

Vertical component of tangential

$$\text{acceleration: } a_{tv} = \frac{3}{2} g \cos^2 \theta$$



Immediately after the stick is released, if this acceleration happens to be larger than g (= acceleration of the block), the block will lag behind the stick.

$$\therefore \frac{3}{2} g \cos^2 \theta_0 \geq g \Rightarrow \cos^2 \theta_0 \geq \frac{2}{3} \Rightarrow \cos \theta_0 \geq \sqrt{\frac{2}{3}}$$

(b) Before the system is released, the block is at rest

$$\therefore \text{Contact force} = mg$$

(c) After release, Contact force = 0

Because acceleration of end A has a vertical component = g

SECTION-II
1. Ans (2.66 to 2.67)

We assume that orbit is circular (though the electron will spiral down into the nucleus). If orbital radius at an instant is r then equating the electrostatic force to the centripetal force we get

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \Rightarrow a = \frac{v^2}{r} = \frac{e^2}{4\pi\epsilon_0 mr} \dots(i)$$

and $v = \sqrt{\frac{e^2}{4\pi\epsilon_0 r}}$ Kinetic energy

$$E = \frac{1}{2}mv^2 = \frac{1}{2} \cdot \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} = \frac{e^2}{8\pi\epsilon_0 r} \dots(ii)$$

Time period of circular motion is $T = \frac{2\pi r}{v}$

Energy loss in one revolution is $\Delta E \simeq \left(\frac{dE}{dt} \right) T$

$$= \frac{e^2 a^2}{6\pi\epsilon_0 c^3} \cdot \frac{2\pi r}{v} = \frac{e^2 r}{3\epsilon_0 c^3 v} \left(\frac{v^2}{r} \right)$$

$$\left[\because a = \frac{v^2}{r} \right]$$

$$= \frac{e^2 v^3}{3\epsilon_0 c^3 r}$$

$$\frac{\Delta E}{E} = \frac{\Delta E}{\frac{1}{2}mv^2} = \frac{e^2 v^3}{3\epsilon_0 c^3 r} \times \frac{8\pi\epsilon_0 r}{e^2} \text{ [using (ii)]}$$

$$= \frac{8\pi}{3} \left(\frac{v}{c} \right)^3 \text{ Since } v \ll c \therefore \frac{\Delta E}{E} \ll 1$$

This means that fraction of energy lost per revolution is small. And because of this reason the radius of the circular path is changing slowly. We can assume the path to be circular for a small interval of time.

2. Ans (25.00 to 25.20)

Threshold wavelength $\lambda_{th} = 240 \text{ nm}$

$$\therefore \text{Work function, } \phi = \frac{hc}{\lambda_{th}} \\ = \frac{(4 \times 10^{-15} \text{ eV s})(3 \times 10^8 \text{ ms}^{-1})}{240 \times 10^{-9}} = 5 \text{ eV}$$

With light of wavelength $2\lambda_0$, the maximum kinetic energy of emitted electrons is 10 eV.

$$\therefore \text{Energy of photon of wavelength } 2\lambda_0 \text{ is } = 10 + 5 \\ = 15 \text{ eV}$$

$$\therefore \text{Energy of photon of wavelength } \lambda_0 \text{ is } = 15 \times 2 \\ = 30 \text{ eV}$$

With light of wavelength λ_0 , the maximum kinetic energy of emitted electrons is $= 30 - 5 = 25 \text{ eV}$

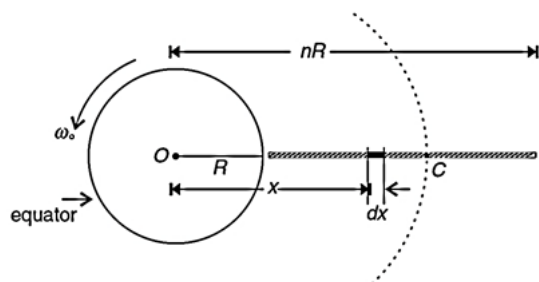
$$\therefore V_0 = 25 \text{ V}$$

3. Ans (12.00)

During each elastic collision the balls will exchange velocities. It is given that all balls are identical (the two colliding balls have no names), we could say that the balls simply pass through each other moving independent of each other. If a ball was at the edge it will take longest time to fall off the other edge.

$$\therefore t_{\max} = \frac{L}{v}$$

4. Ans (2.66 to 2.67)



The rod can stay above the same point only if it rotates with angular speed of earth from west to East (just like geostationary satellites). The necessary centripetal force has to be provided by the gravitational pull of the earth.

Distance of centre of the rod from the centre of the earth is

$$r = \frac{(n-1)R}{2} + R = \frac{(n+1)}{2}R$$

Mass of the rod $m = \lambda (n-1) R$

Where λ = mass per unit length.

Gravitational pull on the rod can be calculated by writing force on an element of length dx .

$$dF = \frac{GM(\lambda dx)}{x^2} \quad [M = \text{mass of earth}]$$

Total force on the rod

$$F = GM\lambda \int_R^{nR} \frac{dx}{x^2} = \frac{GM\lambda}{R} \left[\frac{n-1}{n} \right]$$

$$\therefore F = m\omega_0^2 r$$

$$\frac{GM\lambda}{R} \left[\frac{n-1}{n} \right] = \lambda(n-1)R \cdot \omega_0^2 \left(\frac{n+1}{2} \right) R$$

$$\text{But } M = \frac{4}{3}\pi R^3 \rho$$

$$\therefore \frac{4}{3} \frac{G\pi\rho}{n} = \omega_0^2 \frac{(n+1)}{2}$$

$$\therefore n^2 + n = \frac{8}{3} \frac{\pi G\rho}{\omega_0^2}$$

5. Ans (0.50)

For the right-angle prism, $n = 1.5$, the critical angle

$\alpha = \sin^{-1} \left(\frac{1}{n} \right) = 42^\circ$, which is smaller than the angle of incidence, 45° , at the hypotenuse

of the prism. Therefore total internal reflection occurs, which forms a virtual image. The prism, equivalent to a glass plate of thickness 6 cm, would cause a image shift of

$$\Delta L = 6 \left(1 - \frac{1}{n} \right) = 2 \text{ cm.}$$

Thus the effective object distance for the first lens is

$$u_1 = 10 + 6 + (6 - 2) = 20 \text{ cm.}$$

As $u_1 = f_1$, we have $u_1 = \infty$. Then for the second lens, $u_2 = \infty$.

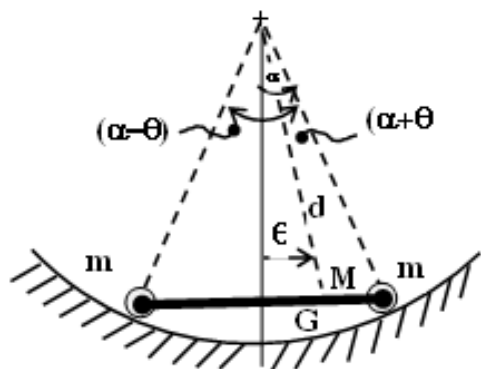
With $f_2 = -10$ cm, we get $u_2 = f_2 = -10$ cm.

Hence the final image is inverted and virtual, 10 cm to the left of the second lens.

The size of the image is

$$l_2 = \left| \frac{f_2}{f_1} \right| 1 = 0.5 \text{ cm}$$

6. Ans (3.00)



$$\alpha = \sin^{-1} \left(\frac{\ell}{2R} \right),$$

$$d = \frac{\ell}{2 \tan \alpha} = \frac{\ell}{\frac{2\ell}{\sqrt{4R^2 - \ell^2}}} = \frac{\sqrt{4R^2 - \ell^2}}{2}$$

$$E = T + V = 2 \left(\frac{1}{2} m v^2 \right) + \frac{1}{2} M v_G^2 + \frac{1}{2} I \dot{\theta}^2 + mgR [\cos \alpha - \cos(\alpha + \theta)] + mgR [\cos \alpha - \cos(\alpha - \theta)] + Mgd [1 - \cos \theta]$$

Set

$$\frac{dE}{dt} = 0, \text{ substitute } v = R\dot{\theta}, v_G = d\dot{\theta}$$

$$\cos \alpha = \frac{\sqrt{4R^2 - \ell^2}}{2R}, \text{ the above expression for}$$

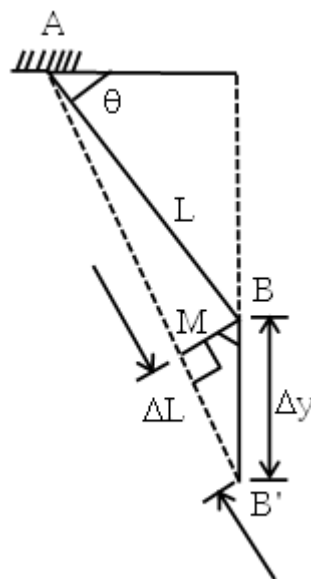
d, and assume small θ to obtain

$$[2mR^2 + MR^2 + \frac{1}{6}M\ell^2]\ddot{\theta} + [g\sqrt{4R^2 - \ell^2} \left(m + \frac{M}{2} \right)]\dot{\theta} = 0$$

$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{2mR^2 + MR^2 + \frac{1}{6}M\ell^2}{g\sqrt{4R^2 - \ell^2} \left(m + \frac{M}{2} \right)}}$$

SECTION-III

1. Ans (8)



Let change in temperature be ΔT

Length of a string change by $\Delta L = L\alpha_2\Delta T$

The wooden plank descends by $\Delta y = \frac{\Delta L}{\sin \theta}$

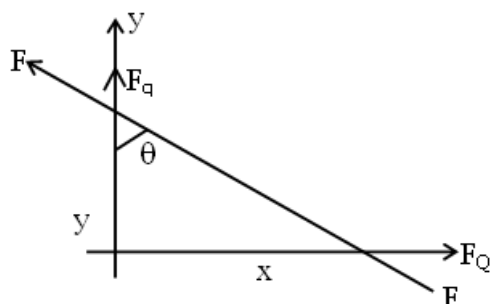
$$\Delta y = 2\Delta L \left[\because \sin \theta = \frac{1}{2} \right]$$

Change in radius of the ball : $\Delta R = R\alpha_1\Delta T$

The centre of the ball will not move if $\Delta y = \Delta R$

$$\Rightarrow 2L\alpha_2\Delta T = R\alpha_1\Delta T \Rightarrow 8R\alpha_2 = R\alpha_1 \Rightarrow \frac{\alpha_1}{\alpha_2} = \frac{8}{1}$$

2. Ans (4)



Force on q along y is

$$F_q = \frac{KqQ}{(x^2 + y^2)} \cdot \cos \theta = \frac{KqQy}{(x^2 + y^2)^{\frac{3}{2}}}$$

Force on Q along x is

$$F_Q = \frac{KqQ}{(x^2 + y^2)} \cdot \sin \theta = \frac{KqQx}{(x^2 + y^2)^{\frac{3}{2}}}$$

$$\therefore \frac{F_q}{F_Q} = \frac{y}{x}$$

$\therefore$ Acceleration of the particles is in the ratio of their respective distance from origin. In the interval q moves a distance r, Q will move through 2r.

3. Ans (9)

$$M\sqrt{2gh} = 3Mv$$

$$\text{for block B, } T - Mg = M \left(\frac{v^2}{\ell} - a \right)$$

$$\text{for block C, } T - Mg = Ma$$

$$a = \frac{v^2}{2\ell} = \frac{2gh}{9.2\ell} = \frac{gh}{9\ell}$$

4. Ans (3)

$$\text{Capacitance of disc system } C = \frac{\epsilon_0 A}{d}$$

When speed of the system is v, the emf induced between two plates is $\epsilon = Bvd$

$$\therefore \text{ Charge on the capacitor } q = C\epsilon = \epsilon_0 ABv$$

$$\text{Energy stored in the capacitor is } U = \frac{1}{2} C \epsilon^2$$

$$\Rightarrow U = \frac{\epsilon_0 A}{2} B^2 v^2 \cdot d$$

Force associated with this energy is

$$F = \frac{dU}{dx} = -\frac{\epsilon_0 A}{2} B^2 d \cdot 2v \frac{dv}{dx}$$

$$= -\epsilon_0 AB^2 \cdot d \cdot v \frac{dv}{dt} \cdot \frac{dt}{dx}$$

$$\text{But } \frac{dt}{dx} = \frac{1}{v} \therefore F = -\epsilon_0 AB^2 d \frac{dv}{dt}$$

Alternatively, this force can be worked out as magnetic force on the rod ($= Bid$) when the current through it is $i = \frac{dq}{dt}$

when the system is displaced by x from its equilibrium and has a speed v: we have

$$M \frac{dv}{dt} = -kx - \epsilon_0 AB^2 \cdot d \frac{dv}{dt}$$

$$\Rightarrow (M + \epsilon_0 AB^2 \cdot d) \frac{dv}{dt} = -kx$$

$$\Rightarrow \frac{dv}{dt} = - \left(\frac{k}{M + \epsilon_0 AB^2 \cdot d} \right) \cdot x$$

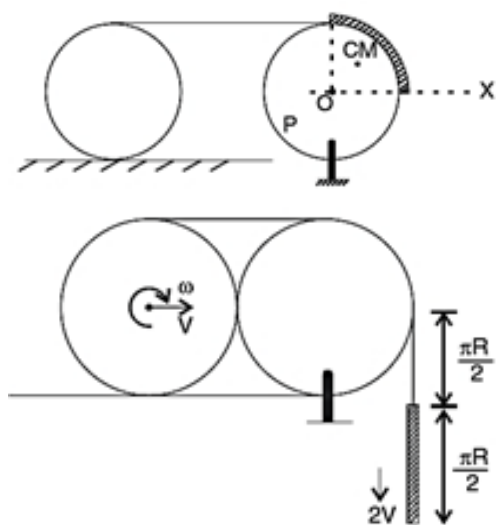
$$\therefore T = 2\pi \sqrt{\frac{M + \epsilon_0 AB^2 \cdot d}{k}}$$

5. Ans (3)

$$r = v \sqrt{\frac{2h}{g}}$$

$$v = r \sqrt{\frac{g}{2h}}$$

6. Ans (1)



As the disc moves through a distance $\frac{\pi R}{2}$, a length of thread $\frac{\pi R}{2}$ gets unwind. Thus a point on the rope moves through a distance πR . Let's apply law of conservation of mechanical energy to the system. Convince yourself that in original position the COM of the rope was at a height of $\frac{2R}{\pi}$ from the line OX.

$$\therefore mg \left[\frac{2R}{\pi} + \frac{3\pi R}{4} \right] = \frac{1}{2}MV^2 + \frac{1}{2} \left(\frac{1}{2}MR^2 \right) \omega^2 + \frac{1}{2}m(2V)^2$$

$$mgR \left(\frac{8 + 3\pi^2}{4\pi} \right) = \frac{3}{4}MV^2 + 2mV^2$$

$$\therefore V^2 = \frac{8 + 3\pi^2}{\pi} \frac{mgR}{(3M + 8m)};$$

$$V = \sqrt{\left(\frac{8 + 3\pi^2}{\pi} \right) \left(\frac{mgR}{3M + 8m} \right)}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (B,C,D)

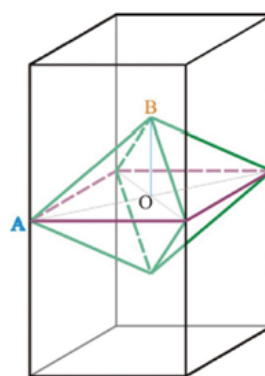
(A) There are four non-regular tetrahedral voids on each face of BCC unit cell (with $\frac{1}{2}$ contribution to the unit cell.) Effective number of tetrahedral voids/unit cell = $6 \times 4 \times \frac{1}{2} = 12$.

Effective number of tetrahedral voids/atom = $12/2 = 6$.

(B) & (C) : A non-regular octahedral void is formed at each edge center (with $\frac{1}{4}$ contribution to unit cell) by two atoms at the corners of the edge (which are nearer to the edge center) and 4 atoms at adjacent body centers (which are slightly farther from the edge center). A non-regular octahedral void is formed at each face center (with $\frac{1}{2}$ contribution to unit cell).

Effective number of octahedral voids per unit cell = $\frac{1}{4} \times 12 + \frac{1}{2} \times 6 = 3 + 3 = 6$.

(D) As the distance $OA > OB$, the atom in the octahedral void at face center (O) touches only the atom at B (body centre).



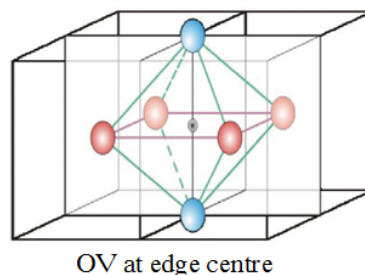
$$OA = \frac{a}{\sqrt{2}};$$

$$OB = \frac{a}{2};$$

$$OB = r + R = \frac{a}{2};$$

$$r + R = \frac{1}{2} \times \frac{4r}{\sqrt{3}};$$

$$\therefore \frac{r}{R} = \left(\frac{2}{\sqrt{3}} - 1 \right) = 0.155$$



OV at edge centre

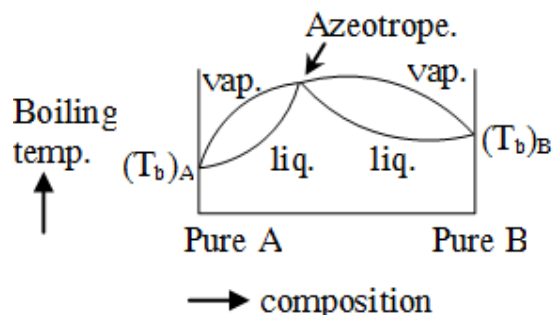
2. Ans (A,C,D)

$\text{Cu}(\text{CN})_2$ = Yellow ppt; $\text{Cu}(\text{OH})_2$ = Blue ppt
 Cu_2O = Red ppt, $\text{Cu}_3[\text{Fe}(\text{CN})_6]_2$ = Green ppt

3. Ans (C,D)

At critical point, no surface of separation exist between liq. & vapour and all properties become identical.

4. Ans (C)



$\Rightarrow$ if mass % of A in liquid = 50%
 Mole ratio of A & B would be 1 : 2
 (Azeotropic composition)

So liquid composition = vapour composition

$\Rightarrow$ If mass % A in liquid > 50%
 (left side of Azeotropic composition)

Mass % of A in vapour > 50%

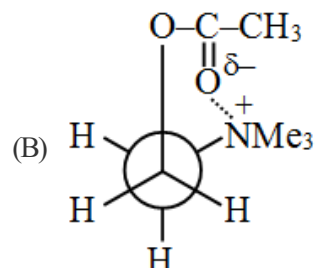
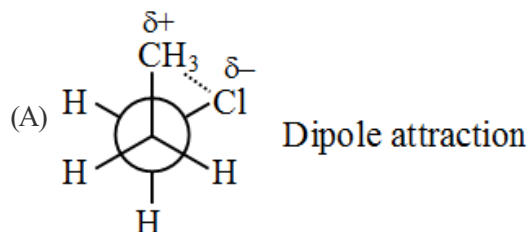
$\Rightarrow$ If mass ratio of A & B in liquid = 1 : 2
 Mole ratio of A & B in liquid = 1 : 4
 (right side of Azeotropic composition)

It can be separated into azeotropic mixture and pure B.

$\Rightarrow$ If mole ratio of A & B is 2 : 1
 (left side of Azeotropic composition)

It can be separated into azeotropic mixture and pure A

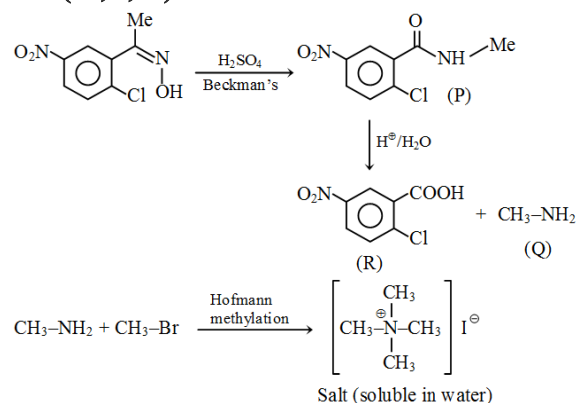
5. Ans (A,B,D)



(C) Anti form is more stable than Gauche form.

(D) H-bonding

6. Ans (A,B,D)



SECTION-II

1. Ans (96.00)

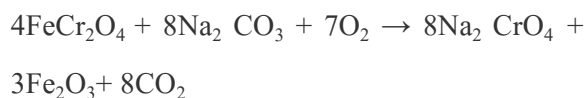
Total heat spent in complete ionisation of 1

mole of H_2A = $0.2 \times 5 + 0.8 \times 15 = 13\text{kJ}$

$\therefore$ heat released during neutralisation = 2 (54.5)

– 13 = 96 kJ

2. **Ans (4.75)**

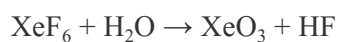
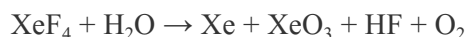
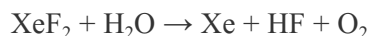


Fe, Cr & O undergo redox change ($\therefore a = 3$)

Moles of O_2 consumed per mole chromite =

$$\frac{7}{4} \quad (\therefore b = 1.75)$$

3. **Ans (16.00)**



$\therefore \text{Xe}$, HF , O_2 & XeO_3 are common in atleast 2 reactions ($\therefore p = 4$)

From given gases, only HCl can be dried using conc. H_2SO_4 ($\therefore q = 4$)

4. **Ans (0.11)**

Let initial concentration of A is $[\text{A}_0]$

$$\text{Rate} = \frac{1}{3} \frac{d[\text{A}]}{dt} = k[\text{A}]^n$$

$$\frac{1}{[\text{A}_t]^{n-1}} - \frac{1}{[\text{A}_0]^{n-1}} = (n-1) 3kt$$

$$\frac{1}{[\text{A}_t]^{n-1}} = (n-1) 3kt + \frac{1}{[\text{A}_0]^{1-n}}$$

According to graph

$$n-1 = 3 \quad n = 4$$

$$\text{Slope} = \tan 45^\circ = 1$$

$$\text{Slope} = (n-1) 3k \quad 1 = 3 \times 3 \times k$$

$$k = \frac{1}{9} \quad \text{Rate} = k[\text{A}]^4$$

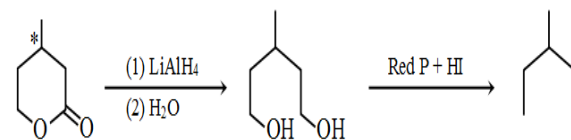
$$= \frac{1}{9} \times [0.1]^4 = \frac{1}{9} \times 10^{-4}$$

5. **Ans (9.00)**

Except (7) all react with $\text{NaNO}_2 + \text{HCl}$.

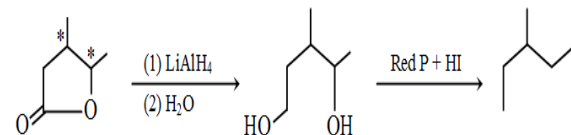
6. **Ans (10.00)**

(1)



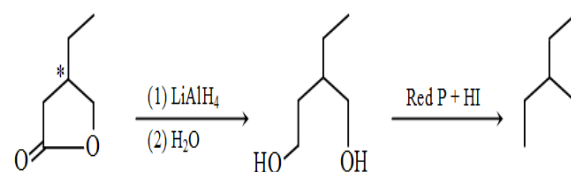
(d/l)

(2)



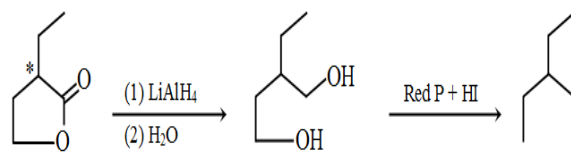
$$(2)^2 = 4$$

(3)



(d/l)

(4)



(d/l)

SECTION-III

1. **Ans (6)**

3d_{zx} orbital

(No radial node $\therefore n - \ell - 1 = 0$ & power of σ is 2 $\Rightarrow \ell = 2$)

$\therefore n = 3 \therefore 3d$. From $\Psi(\theta, \phi)$ orbital is d_{zx} , since on changing to polar co-ordinate system, we get:

$$z = r \cos \theta; x = r \sin \theta \cos \phi; y = r \sin \theta \sin \phi$$

2. **Ans (4)**

ii, iii, iv & vii are correct orders.

3. **Ans (7)**

All other except Aluminothermic process & slag formation.

4. Ans (1)

With NaOH valency factor of $K_xH_y(C_2O_4)_z \cdot nH_2O$ is y

Mili equivalents of salt = mili equivalents of NaOH

$$\frac{2.54}{m} \times 10 \times y = 0.01 \times 30$$

.....(1)

M = molecular weight of salt.

With $KMnO_4$ valency factor of $K_xH_y(C_2O_4)_z \cdot nH_2O$ is $2z$

$$\frac{2.54}{m} \times 10 \times 2z = 0.01 \times 40$$

.....(2)

$$\frac{\text{eq. 1}}{\text{eq. 2}} = \frac{y}{2z} = \frac{3}{4}$$

$$\frac{y}{z} = \frac{3}{2}$$

Apply charge balancing



$$x + y - 2z = 0 \quad (y = 3, z = 2)$$

$$x + 3 - 4 = 0$$

$$x = 1$$



Calculate 'm' from eq. 1

$$m = 254$$

$$39 + 3 + 88 \times 2 + 18n = 254$$

$$n = 2$$

$$\text{Hence } x = 1, y = 3, z = 2$$

$$\text{Ans. } \frac{1 + 3 - 2}{2} = 1$$

5. Ans (6)

(EtONa/EtOH), (CH_3MgBr/Et_2O) , (HBr),
(CH_3COONa/CH_3COOH), ($CH_3OH/dil.HCl$),
($LiAlH_4/THF$) can do it.

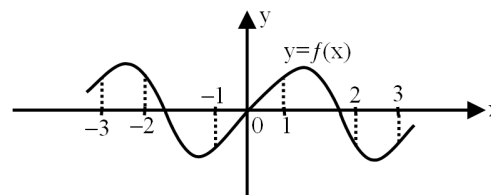
6. Ans (8)

Reactions (1), (2), (3), (4), (5), (6), (8) and (10) proceed through free radical mechanism.

PART-3 : MATHEMATICS

SECTION-I

1. Ans (A,D)



$$f(1) = 1, f(2) = -1, f(3) = -1, f(2) = -1, f(0) = 0$$

$\therefore f(x) = 0$ has at least 3 roots.

$f'(x) = 0$ has atleast four real roots

(By Rolle's theorem, $f'(x) = 0$ must have atleast one root in interval (2, 3) and (-3, -2),

Because $f(2) = f(3)$ & $f(-2) = f(-3)$)

$$\text{Let } g(x) = f'(x)^2 + f''(x)$$

$$e^{f(x)} \cdot g(x) = e^{f(x)} \cdot (f'(x))^2 + e^{f(x)} \cdot f''(x) = [e^{f(x)} \cdot f'(x)]'$$

So $e^{f(x)} \cdot f'(x) = 0$ has atleast four real roots

$\therefore$ By Rolle's theorem, $g(x)$ has atleast three real roots

2. Ans (B,C)

Take dot product with $\vec{a}$ we get

$$(\vec{r} \cdot \vec{b}) = \frac{|\vec{a}|^2}{\vec{a} \cdot \vec{b}}$$

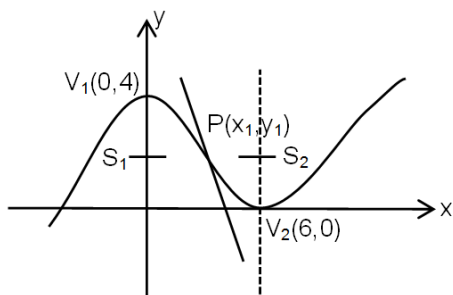
Take cross product with $\vec{b}$ we get

$$\frac{|\vec{a}|^2}{\vec{a} \cdot \vec{b}} \vec{a} - (\vec{a} \cdot \vec{b}) \vec{r} = (\vec{b} \times \vec{a})$$

$$\Rightarrow \vec{r} = \frac{|\vec{a}|^2}{(\vec{a} \cdot \vec{b})^2} \vec{a} + \frac{\vec{a} \times \vec{b}}{\vec{a} \cdot \vec{b}}$$

$$\Rightarrow f(\theta) = [\vec{r} \cdot \vec{a} \cdot \vec{b}] = \frac{|\vec{a} \times \vec{b}|^2}{\vec{a} \cdot \vec{b}} = \frac{\sin^2 \theta}{\cos \theta}$$

3. Ans (A,B,D)



For the parabolas to touch, they will have their concavities in opposite direction & axis parallel to y-axis. Let, $P_1 : x^2 = \lambda (y - 4)$

$$P_2 : (x - 6)^2 = \mu y.$$

As they touch in one point, solving the equations, we get :

$$x^2 = \lambda$$

$$\left[\frac{(x-6)^2}{\mu} - 4 \right]$$

$$\Rightarrow (\lambda - \mu) x^2 - 12\lambda x + 36\lambda - 4\lambda\mu = 0$$

$$D = 0 \Rightarrow 36\lambda^2 - (\lambda - \mu)(36\lambda - 4\lambda\mu) = 0$$

$$\Rightarrow \mu - \lambda = 9$$

$$\text{If } P_1 \text{ and } P_2 \text{ are equal then, } P_1 : x^2 = -\frac{9}{2}(y - 4)$$

$$\text{and } P_2 : (x - 6)^2 = \frac{9}{2}y$$

$$\text{L.R.} = \frac{9}{2}$$

Let them meet at $P(x_1, y_1)$.

Then, $\frac{dy}{dx} \Big|_{(x_1, y_1)}$ will be same for both

$$\therefore -\frac{2x_1}{\lambda} = \frac{2(x_2 - 6)}{\mu} \Rightarrow x_1 = 3 \Rightarrow y_1 = 2$$

$$\therefore P \equiv (3, 2)$$

4. Ans (A,C,D)

$$(1 + x + x^2)^{2014} = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6 + \dots$$

Substituting $-1, -\omega, -\omega^2$ where $\omega = e^{i2\pi/3}$

$$1 = a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + a_6 \dots \text{ (i)}$$

$$(1 - \omega + \omega^2)^{2014} = a_0 - a_1\omega + a_2\omega^2 - a_3 + a_4\omega - a_5\omega^2 + a_6 \dots \text{ (ii)}$$

$$(1 - \omega^2 + \omega)^{2014} = a_0 - a_1\omega^2 + a_2\omega - a_3 + a_4\omega^2 - a_5\omega + a_6 \dots \text{ (iii)}$$

$$(i) + (ii) + (iii) \Rightarrow a_0 - a_3 + a_6 \dots =$$

$$\frac{1 + 2^{2014}(\omega + \omega^2)}{3} = \frac{1 - 2^{2014}}{3}$$

$$(i) + \omega^2(ii) + \omega(iii) \Rightarrow a_1 - a_4 + a_7 \dots = -\left(\frac{1 + 2^{2015}}{3}\right)$$

$$(i) + \omega(ii) + \omega^2(iii) \Rightarrow a_2 - a_5 + a_8 \dots = \frac{1 - 2^{2014}}{3} \Rightarrow |A| = |C| < |B|$$

5. Ans (A,C)

$$AP + PB = AB$$

$$\Rightarrow \sqrt{(r_1+r)^2 - (r_1-r)^2} + \sqrt{(r_2+r)^2 - (r_2-r)^2} = \sqrt{(r_1+r_2)^2 - (r_1-r_2)^2}$$

$$\Rightarrow 2\sqrt{r_1r_2} + 2\sqrt{r_1r_2} = 2\sqrt{r_1r_2} \Rightarrow rr_1 + rr_2 + 2r$$

$$\sqrt{r_1r_2} = r_1r_2 \dots (1)$$

$$\Rightarrow \frac{1}{\sqrt{r_1}} + \frac{1}{\sqrt{r_2}} = \frac{1}{\sqrt{r}} \Rightarrow \sqrt{r_1}, 2\sqrt{r}, \sqrt{r_2}$$

are in H.P.

Apply $GM \geq HM$

$$\Rightarrow (2\sqrt{r})^2 < \sqrt{r_1} \sqrt{r_2} \Rightarrow 4r < \sqrt{r_1r_2} \Rightarrow 2\sqrt{r_1r_2} > 8r$$

$$\Rightarrow 2\sqrt{r_1r_2} - r > 7r > 0 \dots (2)$$

$$\text{Now } (C_1C_2)^2 - (C_1C)^2 - (C_2C)^2$$

$$= r_1^2 + r_2^2 + 2r_1r_2 - r^2 - r_1^2 - 2rr_1 - r^2 - r_2^2 - 2rr_2$$

$$= -2rr_1 - 2rr_2 + 2r_1r_2 - 2r^2 = 4r \sqrt{r_1r_2} - 2r^2$$

$$= 2r (2\sqrt{r_1r_2} - r) = \text{positive} \Rightarrow \Delta C_1CC_2$$

is obtuse angle triangle

If $r_1 = r_2$ then from equation (1), $r_1 = 4r$

$$\text{and } \sin \frac{A}{2} = \frac{r_1}{r + r_1}$$

6. **Ans (A,B,C,D)**

Let $(5,3,-3)$ is A, $(0,-1,-1)$ is B, $(0,5,11)$ is C and $(5,9,9)$ is D.

New ABCD is rectangle and diagonal BD has direction cosines as

$$\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$$

It means the vertices whose position is changed are vertices A and C

Let foot of A on BD is F $(\lambda, -1+2\lambda, -1+2\lambda)$

Because AF perpendicular to BD, hence $1(\lambda - 5) + 2(2\lambda - 4) + 2(2\lambda + 2) = 0$

$$\Rightarrow \lambda = 1$$

$\Rightarrow$ F is $(1,1,1)$

Let mid point of BD is E $= \left(\frac{5}{2}, 4, 4\right)$

Normal vector to plane ABCD is $\overrightarrow{AD} \times \overrightarrow{CD}$

$$= (6\hat{j} + 12\hat{k}) \times (5\hat{i} + 4\hat{j} - 2\hat{k})$$

$$= -60\hat{i} + 60\hat{j} - 30\hat{k}$$

$$\text{or } -\frac{2}{3}\hat{i} + \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$$

Now distance between A and F is 6

$\Rightarrow$ New position of A is

$$\left(1 - \frac{2}{3} \times 6, 1 + \frac{2}{3} \times 6, 1 - \frac{1}{3} \times 6\right) \quad \text{or}$$

$$\left(1 + \frac{2}{3} \times 6, 1 - \frac{2}{3} \times 6, 1 + \frac{1}{3} \times 6\right)$$

$\Rightarrow$ New A is either $(-3,5,-1)$ or $(5,-3,3)$

If new A is $(-3,5,-1)$ then new C is $(8,3,9)$ and

if new A is $(5,-3,3)$ then new C is $(0,11,5)$

(calculate new C by applying mid point formula)

SECTION-II

1. **Ans (1275.00)**

$$f(x) = \prod_{n=1}^{50} (x - n)^{n(51-n)}$$

$$\ln(f(x)) = \sum_{n=1}^{50} n(51-n) \ln(x-n)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \sum_{n=1}^{50} \frac{n(51-n)}{x-n}$$

$$\Rightarrow \frac{f'(51)}{f(51)} = 1275$$

2. **Ans (960.00)**

A

B

$$\begin{bmatrix} 4\lambda_1 + 1 & 4\lambda_2 + 1 \\ 4\lambda_3 + 1 & 4\lambda_4 + 1 \end{bmatrix} \times \begin{bmatrix} \text{---} & \text{---} \\ \text{---} & \text{---} \end{bmatrix}$$

$$= \begin{bmatrix} 4\lambda & 4\lambda \\ 4\lambda & 4\lambda \end{bmatrix}$$

$\Rightarrow$ B can be of the form

$$\begin{bmatrix} 4\lambda_5 - 1 & 4\lambda_6 - 1 \\ 4\lambda_7 + 1 & 4\lambda_8 + 1 \end{bmatrix} \quad \text{Interchange possible}$$

$$\Rightarrow 4 \times {}^5P_2 \times {}^4P_2 = 4 \times 20 \times 12 = 960$$

3. Ans (12.00)

Now,

$$\frac{\sum_{r=0}^4 {}^4C_r \sin 2rx}{\sum_{r=0}^4 {}^4C_r \cos 2rx} = \frac{\sum_{r=0}^4 {}^4C_{4-r} \sin 2(4-r)x}{\sum_{r=0}^4 {}^4C_{4-r} \cos 2(4-r)x}$$

$$= \frac{\sum_{r=0}^4 {}^4C_r [\sin 2rx + \sin 2(4-r)x]}{\sum_{r=0}^4 {}^4C_r [\cos 2rx + \cos 2(4-r)x]}$$

$$= \frac{\sum_{r=0}^4 {}^4C_r \cdot 2 \sin 4x \cdot \cos(2rx - 4x)}{\sum_{r=0}^4 {}^4C_r \cdot 2 \cos 4x \cdot \cos(2rx - 4x)} = \tan 4x$$

$$\left(\because \frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d} \right)$$

4. Ans (2.00)

$z_1, z_2 \dots z_9$ are roots of equation $z^{10} = 1$ other than $z = 1$

$\Rightarrow$ all are vertices of a regular decagon

$$\Rightarrow z_1^r + z_2^r + \dots + z_9^r = \begin{cases} -1 & ; r \neq 10\lambda \\ 9 & ; r = 10\lambda \end{cases} \quad \forall \lambda \in \mathbb{N}$$

$$\Rightarrow \sum_{r=0}^9 \sum_{i=1}^9 z_i^r = 36 - 34 = 2$$

5. Ans (324.00)

Number of ways of discontinuity at a point $x = c$ are 3.

(A) LHL at $x = c$ is equal to $f(c)$ but not equal to RHL at $x = c$

(B) LHL at $x = c$ is not equal to $f(c)$ but equal to RHL at $x = c$

(C) LHL at $x = c$ is neither equal to $f(c)$ nor equal to RHL at $x = c$ (where $f(c)$) is equal to RHL at $x = c$

6. Ans (12.80)

5 is of the form $8\lambda - 3, \lambda \in \mathbb{I}$

11 is of the form $8\lambda + 3, \lambda \in \mathbb{I}$

13 is of the form $8\lambda - 3, \lambda \in \mathbb{I}$

19 is of the form $8\lambda + 3, \lambda \in \mathbb{I}$

Now If we multiply the numbers of the forms

(i) $(8\lambda + 3)(8\lambda + 3)$ we get number of the form $8\lambda + 1$

(ii) $(8\lambda - 3)(8\lambda - 3)$ we get number of the form $8\lambda + 1$

(iii) $(8\lambda + 3)(8\lambda - 3)$ we get number of the form $8\lambda - 1$

(iv) $(8\lambda - 1)(8\lambda - 1)$ we get number of the form $8\lambda + 1$

(v) $(8\lambda + 1)(8\lambda - 1)$ we get number of the form $8\lambda - 1$

(vi) $(8\lambda - 1)(8\lambda - 1)$ we get number of the form $8\lambda + 1$

$\Rightarrow$ for making factor of the form $8\lambda + 1$ we have to select 5 and 13 such that their sum power becomes even, as well as we have to select 11 and 19 such that their sum power becomes even.

$\Rightarrow$ powers of 5 and 13 respectively can be equal to (0, 0), (0, 2), (1, 1), (1, 3), (2, 0), (2, 2), (3, 1), (3, 3) and powers of 11 and 19 respectively can be equal to (0, 0), (0, 2), (1, 1), (1, 3), (2, 0), (2, 2), (3, 1), (3, 3)

$\Rightarrow$ total number of divisors of N in the form of $8\lambda + 1$ are equal to $8 \times 8 = 64 = a$

$$\Rightarrow \frac{a}{5} = 12.8$$

SECTION-III

1. Ans (3)

$$\int_0^1 f'(x) dx = f(1) - f(0) = 1$$

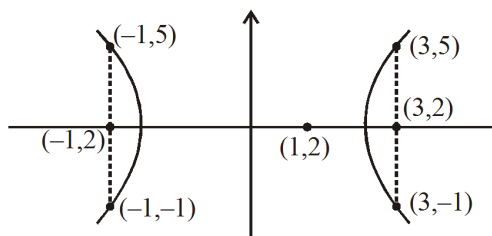
$$= \int_0^1 (1+x^2)^{-1/4} (1+x^2)^{1/4} f'(x) dx$$

(Cauchy Schwarz inequality)

$$\leq \left(\int_0^1 (1+x^2)^{-1/2} dx \right)^{1/2} \left(\int_0^1 (1+x^2)^{1/2} (f'(x))^2 dx \right)^{1/2}$$

$$\Rightarrow \int_0^1 (1+x^2)^{1/2} (f'(x))^2 dx \geq \frac{1}{\ln(1+\sqrt{2})}$$

2. Ans (4)



$$ae = 2,$$

$$\frac{b^2}{a} = 3$$

$$a(e^2 - 1) = 3$$

$$\frac{2}{e}(e^2 - 1) = 3$$

$$2e^2 - 3e - 2 = 0$$

$$2e^2 - 4e + e - 2 = 0$$

$$\Rightarrow e = 2 \text{ } (-1/2 \text{ rejected})$$

$$\text{foci } (3, 2), (-1, 2) \Rightarrow \text{center } (1, 2)$$

$$\frac{a}{e} = \frac{1}{2} \Rightarrow \text{directrix are } x = \frac{3}{2} \text{ and } x = \frac{1}{2}$$

3. Ans (4)

$$\text{Let, Arg } (z) = \theta \in \left[0, \frac{\pi}{2}\right]$$

$$\Rightarrow \text{Arg } \frac{1}{z} = -\theta$$

$$\text{Area of parallelogram} = \frac{12}{13} = |z| \left| \frac{1}{z} \right| \sin 2\theta$$

$$\Rightarrow \sin 2\theta = \frac{12}{13}$$

$$\text{Now, } \left| z + \frac{1}{z} \right|^2 = \left(z + \frac{1}{z} \right) \left(\bar{z} + \frac{1}{\bar{z}} \right)$$

$$= |z|^2 + \frac{z}{\bar{z}} + \frac{\bar{z}}{z} + \frac{1}{|z|^2}$$

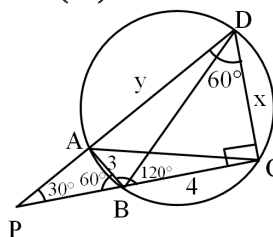
$$= |z|^2 + \frac{1}{|z|^2} + 2 \cos 2\theta$$

$$\therefore \left| z + \frac{1}{z} \right|^2 \geq 2 + 2 \cos 2\theta$$

Equality will hold when $|z| = 1$

$$\therefore \left| z + \frac{1}{z} \right|_{\min}^2 = 2 + 2 \cos 2\theta$$

4. Ans (1)



$$PB = \frac{3}{\sin 30^\circ} = 6$$

$$\therefore \text{In } \triangle PCD \Rightarrow x = 10 \tan 30^\circ = \left(\frac{10\sqrt{3}}{3} \right)$$

5. Ans (3)

$$(7 + 4\sqrt{3})^n = a_n + b_n \sqrt{3} \Rightarrow (7 - 4\sqrt{3})^n = a_n - b_n \sqrt{3}$$

$$\Rightarrow a_n = \frac{1}{2} [(7 + 4\sqrt{3})^n + (7 - 4\sqrt{3})^n] \text{ and } b_n = \frac{1}{2\sqrt{3}} [(7 + 4\sqrt{3})^n - (7 - 4\sqrt{3})^n]$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \sqrt{3} \Rightarrow \left(\lim_{n \rightarrow \infty} \frac{a_n}{b_n} \right)^2 = 3$$

6. Ans (8)

$$\sec \theta + \operatorname{cosec} \theta = c \Rightarrow \sin \theta + \cos \theta = c \sin \theta \cos \theta$$

sin

$$\left(\theta + \frac{\pi}{4} \right) = \frac{c}{2\sqrt{2}} \sin 2\theta \text{ (where } \sin \theta \neq 0, \cos \theta \neq 0)$$

$$\text{graph of } y = \sin \left(\theta + \frac{\pi}{4} \right) \text{ and } y = \frac{c}{2\sqrt{2}} \sin 2\theta \text{ intersect}$$

$$\text{at two point if } \frac{c}{2\sqrt{2}} \in (-1, 1)$$